Can AI write a shop poster in Hindi and Bengali?

Product case study · Swapnil Ghatak · October 2026

Summary

I wanted to know whether today's image models can make a festive poster that a small shop owner could post straight away, with the shop's message spelt correctly in Hindi or Bengali. I gave three models the same 6 prompts (3 Hindi, 3 Bengali, from a two-word greeting up to prices and phone numbers) and asked 11 people to rate the results without knowing which model made which poster. That gave 153 poster ratings and 51 "best poster" votes. I also went through every poster letter by letter.

Gemini 3.1 Flash Image Preview came out well ahead. Raters marked its text exactly right 90% of the time and never marked it wrong, and it was picked as the best poster in 63% of votes. In my own check, all 6 of its posters were correct.

GPT Image 1 was close on simple text but kept making small mistakes: a missing nukta, ষ written as ব, English digits where I asked for Bengali ones. Raters said they'd post only 33% of its posters as they were. Gemini 2.5 Flash Image struggled the most: 59% of its posters were rated wrong or unreadable, and in Bengali that went up to 86%. None of its 6 posters was fully correct.

Two things surprised me. First, nobody wanted to post a poster with wrong text (0%), so text accuracy works like a pass/fail gate before design even matters. Second, who does the rating matters a lot. People who read only Hindi caught every wrong Hindi poster, but people who said they read both scripts let about half of those through as "exactly correct".

1. What I tested and why

The category is marketing and commerce. The use case is a small local business, like a sweet shop, saree store, mobile repair shop or restaurant, that wants a festival or offer poster in the local language to share on WhatsApp or Instagram, without paying a designer.

I picked this because it's a very Indian problem. Over 7.8 crore businesses are registered on the government's Udyam platforms (Ministry of MSME, March 2026), and most of their customers read Hindi, Bengali or another regional script. Festivals like Diwali and Durga Puja are when they sell the most. If the poster has a spelling mistake in the greeting or the wrong price, it's worse than having no poster at all.

It also matters to AI labs. Writing text inside an image is one of the harder things for image models, and Indic scripts make it harder still, because vowel signs attach to letters, letters join into conjuncts (প্র, ক্ষ, র্ক) and some scripts have their own digits. Most public benchmarks only test English text. If a lab wants Indian users to rely on its model for real work, it needs to know how often the text comes out exactly right in each script, and which characters break.

I judged the posters on three things, in this order:

1. Is the text exactly right? Every character as asked, nothing extra. If this fails, nothing else matters to the shop owner.
2. Would you post it as-is? A simple yes/no that covers layout, readability and whether it feels right for the festival.
3. Which poster is best overall? The same kind of preference vote used on LMArena.

I left out English text, photorealism, art style and editing, to keep the test narrow.

2. How I ran it

Prompts

All 6 prompts used the same template so every model got identical instructions. They get harder as they go: Easy is a two-word greeting, Medium is a phrase with conjuncts or a nukta, and Hard is two lines with a price, a phone number or Bengali digits.

ID	Script	Difficulty	Business / occasion	Exact text required
P01	Hindi	Easy	Sweet shop, Varanasi, Diwali	शुभ दीपावली
P02	Hindi	Medium	Saree shop, Lucknow, festive season	त्योहार की ख़ास पेशकश
P03	Hindi	Hard	Mobile repair shop, Patna (2 lines incl. price + phone)	मोबाइल रिपेयर – सिर्फ़ ₹199 से शुरू / संपर्क करें: 98765 43210
P05	Bengali	Easy	Sweet shop, Kolkata, Durga Puja	শুভ শারদীয়া
P06	Bengali	Medium	Saree shop, Kolkata, Durga Puja	পুজোর বিশেষ ছাড়
P07	Bengali	Hard	Restaurant, Siliguri, Durga Puja (price in Bengali numerals)	পুজো স্পেশাল থালি – মাত্র ₹২৪৯

Each prompt started with "Create an image: A square promotional poster for ..." and ended with "... No other text." The full prompts are in Appendix C.

Models and where I generated the images

Company	Model	Where generated	Model name shown on screen	Settings
Google	Gemini 3.1 Flash Image Preview	Gemini app (gemini.google.com), free tier	The app doesn't show a model ID. Its image model is "Nano Banana 2", which Google identifies as Gemini 3.1 Flash Image	Default, new chat per prompt
Google	Gemini 2.5 Flash Image	arena.ai, Direct mode, Image	`gemini-2.5-flash-image-preview (nano-banana)`	Default, new chat per prompt
OpenAI	GPT Image 1	arena.ai, Direct mode, Image	`gpt-image-1`	Default, new chat per prompt

To keep it fair, I used exactly the same prompt for every model, kept the first image each time without regenerating, and took a screenshot of every generation showing the prompt and the result. Everything was generated on 11 October 2026.

A few things didn't go to plan, so I'm noting them here. I first tried the official APIs, but all three needed paid credits, so I switched to the free interfaces and documented the workflow. On arena.ai, my first try for P02 accidentally ran in Battle mode and gave me two other models (flux-2-pro and a Nano Banana 2.1 version), so I threw it away and redid it with the right model. The Gemini app gave 2048-pixel images while the others were 1024, so I resized all 18 to 1024×1024 before rating, so sharpness wouldn't give away the model. And in the Gemini app I pasted "Create an image:" twice for P01 by mistake; the rest of that prompt was the same.

Raters and how they judged

I shared a Google Form with friends and classmates. 12 people filled it in. I removed one response, my own test run, because I already knew which model made which poster. That left 11 raters aged 21 to 27. 10 rated the Hindi posters and 7 rated the Bengali posters (6 did both), and 3 of them own or help run a shop or small business. Everyone ticked the consent statement and gave their name, email and age (Appendix A).

My first version of the form took about 10 minutes and felt too long, so I cut it down to about 4. Each person picked the script they read and only saw those posters, which also meant text was only judged by people who could read it. For each set, the form showed the exact text the poster should have, then the three posters as A, B and C, with model names hidden and the order changed from set to set. Raters answered three quick questions: is the text on each poster exactly right, slightly wrong or wrong/unreadable; would you post each one as-is; and which one is best.

Separately, I checked all 18 posters letter by letter (I read both scripts), so I'd have a reliable reference to compare the raters against.

3. What I found

Overall results

#	Model	Company	Text exactly correct (raters)	Text wrong / unreadable (raters)	"Would post as-is"	Picked best overall	Exactly correct (my audit)
1	Gemini 3.1 Flash Image Preview	Google	90%	0%	71%	32/51 (63%)	6/6
2	GPT Image 1	OpenAI	65%	2%	33%	9/51 (18%)	3/6
3	Gemini 2.5 Flash Image	Google	22%	59%	22%	10/51 (20%)	0/6

Rater percentages are out of 51 ratings per model (10 Hindi raters × 3 sets + 7 Bengali raters × 3 sets). "Picked best overall" is out of 51 votes. "My audit" is my own letter-by-letter check of each model's 6 posters.

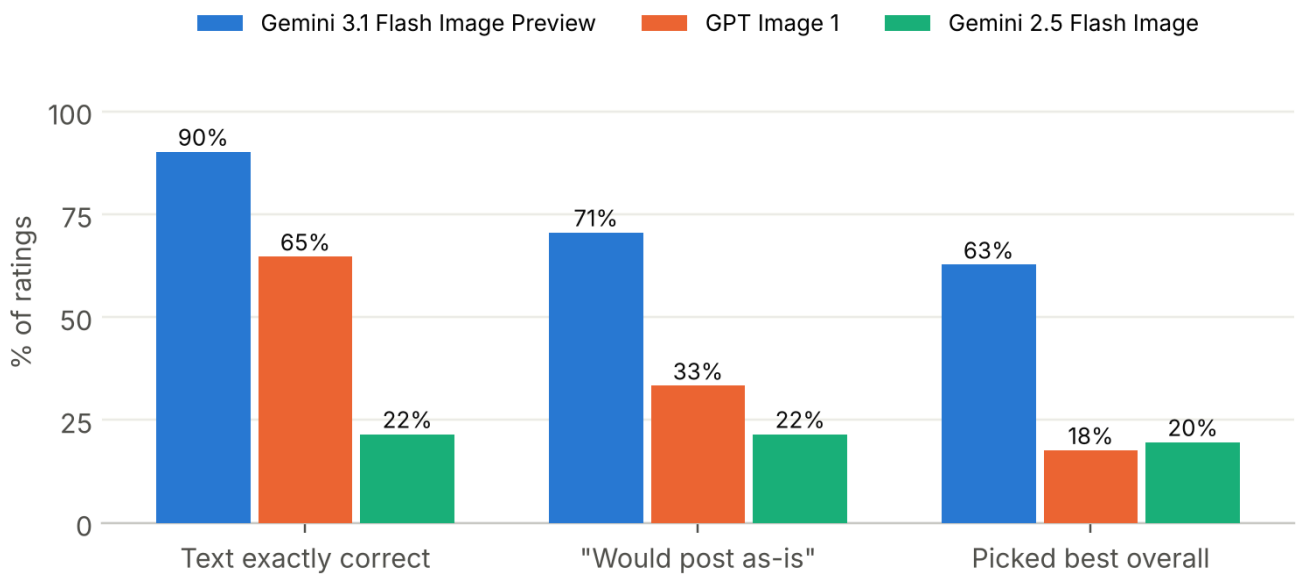


Figure 1. The order is the same on all three measures. GPT Image 1 is much closer to Gemini 3.1 on text accuracy than on whether people would post it or prefer it.

Bengali is where the models split

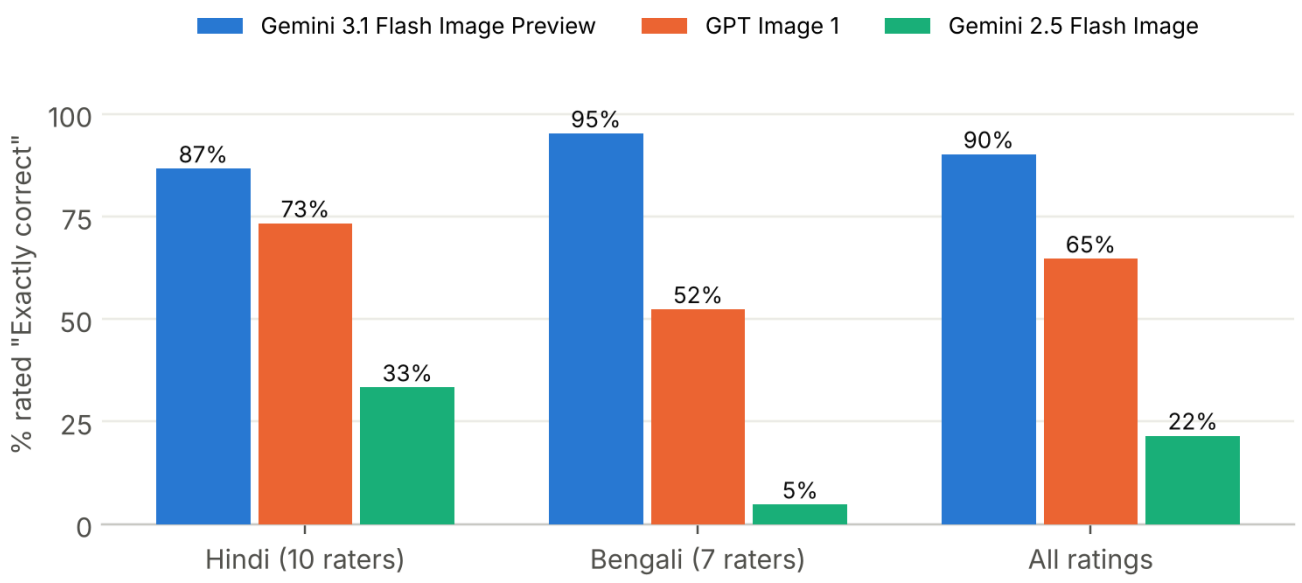


Figure 2. Share of ratings marked "Exactly correct", by script.

Gemini 3.1 did just as well in Bengali (95%) as in Hindi (87%). The other two dropped a lot: GPT Image 1 went from 73% in Hindi to 52% in Bengali, and Gemini 2.5 from 33% to 5%. Hindi probably has the most training data of any Indian language, so a model can look fine on Hindi and still fall apart on the next scripts. If I'd only tested Hindi, I would have made GPT Image 1 look much more ready than it is.

Harder text trips up GPT Image 1 but not Gemini 3.1

Model	Easy	Medium	Hard	Hindi	Bengali
Gemini 3.1 Flash Image Preview	82%	94%	94%	87%	95%
GPT Image 1	82%	59%	53%	73%	52%
Gemini 2.5 Flash Image	24%	12%	29%	33%	5%

Share of ratings marked "Exactly correct", by difficulty and by script.

On the two-word greetings, GPT Image 1 and Gemini 3.1 were level (82% each). Once the text got longer, with prices, phone numbers and conjuncts, GPT Image 1 fell to 53% while Gemini 3.1 stayed at 94%. Its mistakes were exactly the kind a shop owner would care about: the nukta missing in सिर्फ़, ष turned into ব in বিশেষ ("special"), and the price in English digits when I asked for Bengali ones.

Correct text is needed, but it isn't enough

Raters didn't want to post a single poster they thought had wrong text. So correct text is the minimum. But it doesn't win on its own: GPT Image 1's text was rated exactly right 65% of the time, yet people would post only 33% of its posters and chose it as best in just 18% of votes. Looking at the posters, its designs were plainer (P03 is white text on a flat red background), while Gemini 3.1 made fuller, more festive scenes. So a lab tracking Indic text should look at a usability measure alongside it.

Who rates matters as much as what's rated

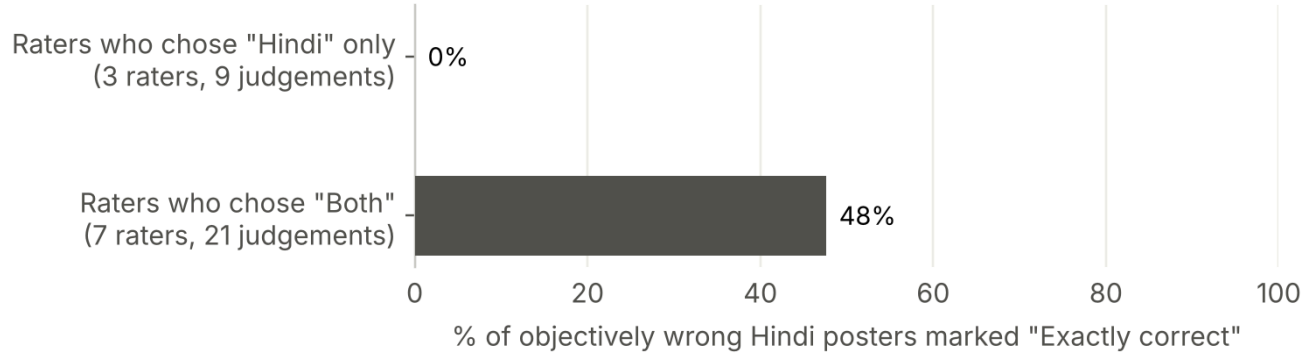


Figure 3. Hindi posters I found to be wrong, and how often each group of raters still called them "Exactly correct".

Raters agreed with my letter-by-letter check 73% of the time: 64% for Hindi and 84% for Bengali. Most of that gap comes from who rated the Hindi posters. The three people who picked "Hindi" only caught every wrong Hindi poster. The seven who said they read both scripts marked 48% of the wrong Hindi posters as exactly correct, for example accepting "दोपावली" for "दीपावली". Small mistakes were hard for everyone: raters matched my check on only 54% of posters that had one. For a bigger version of this eval, that means text should only be judged by native readers of that script, and they should be tested with posters where the answer is already known.

Examples of what went wrong



Gemini 2.5, P01. Asked for শুভ দীপাবলী. Wrote "दोपावली" and added text nobody asked for (वारानशी मिश्रान बंनदार).



Gemini 2.5, P06. Asked for পুজোর বিশেষ ছাড়. Got "পুজার শিশস ছার্দ", which doesn't mean anything.



Gemini 2.5, P07. থালি became খালী, and the price changed from ₹২৪৯ to ₹২১৭. A shop would be advertising the wrong price.



GPT Image 1, P06. Looks clean, but বিশেষ is written বিশেব. One letter off. 5 of 7 raters spotted it.



GPT Image 1, P07. The price is in English digits (249), though I asked for Bengali ones (২৪৯).



Gemini 3.1, P07. All correct, including the conjunct স্পে and the Bengali digits ₹২৪৯.

The kinds of mistakes I saw: wrong vowel sign (ী → ো), broken or missing conjuncts (মোবাইল → মোবাাল, संपर्क → संपक), one letter swapped for another (ষ → ব, থ → খ), a missing nukta (सिर्फ़ → सिर्फ), digits in the wrong script (২৪৯ → 249), a wrong number (₹২৪৯ → ₹২১৭), and extra text that wasn't asked for. Grouping errors like this is more useful to a model team than a single pass/fail score, because it tells them what to fix.

Limitations

- It's a small pilot: 6 prompts, 11 raters and one image per prompt per model. The order of the models held up in every cut of the data, but the exact percentages could move a fair bit with more data.
- Within a set, every rater saw the posters in the same A/B/C order (I only shuffled between sets), so there could be some position bias. Gemini 3.1 also won a set where it was poster C (P02), so I don't think it changed the result, but next time I'd shuffle per person.
- Because I used the free interfaces, I couldn't fix settings like aspect ratio, and the Gemini app doesn't show its model ID. The Gemini app also adds a small sparkle watermark, which a sharp-eyed rater might have noticed.
- Most raters were MBA students rather than shop owners (3 of 11 run a business), and most Hindi ratings came from people who read both scripts, who missed more errors.

4. How I'd scale it

The goal would be a regularly updated leaderboard for Indic text in images, covering all 22 scheduled languages, with results a lab can act on. Here's what I'd do:

- Build a prompt bank of around 50 prompts per language, based on what small businesses actually need (festival greetings, discount offers, price lists, shop signs, opening hours). Each prompt would be tagged with what it tests, like conjuncts, vowel signs, nukta, native digits or two-line layouts, and written by native speakers rather than machine-translated.
- Use only native readers of each script as raters. Before rating, each person passes a short test on posters with known mistakes, and about 1 in 10 posters in every task is a hidden test poster. People who keep missing those get retrained or removed. Platforms that already run large networks of Indian-language contributors are a natural place to find them.
- Keep costs down with two stages. An OCR check flags obvious failures automatically (like a wrong price), and people judge the rest plus the design questions. OCR isn't reliable on decorative festive fonts, so a person should always make the final call.
- Improve the rating task: shuffle A/B/C for each rater, get 3 ratings per poster, and ask which character was wrong instead of just "right or wrong".
- Report the numbers with sample sizes and confidence ranges: text accuracy per script, the "would post" rate, and an Elo-style preference score like LMArena's.
- Add new models as they come out by running them through the full prompt bank through their APIs, and publish per-script error breakdowns that model teams can use.
- Every so often, bring in real shop owners from each region for the "would you post this?" question, since they're the people this eval is really about.

5. Dashboard and rating screen

I made a clickable prototype using the real data from this pilot: the `dashboard/` folder of this repository (live link in the README). It has three parts. The leaderboard ranks models and lets you filter by script, difficulty and measure, with the sample size next to every number so nobody reads too much into a small cell. The failure explorer shows every poster with the text it should have had, what raters said, and the type of mistake, which is what a model team would use to debug. The rating screen is the phone view a rater sees: the required text, three unnamed posters and three taps, the same flow I used here.

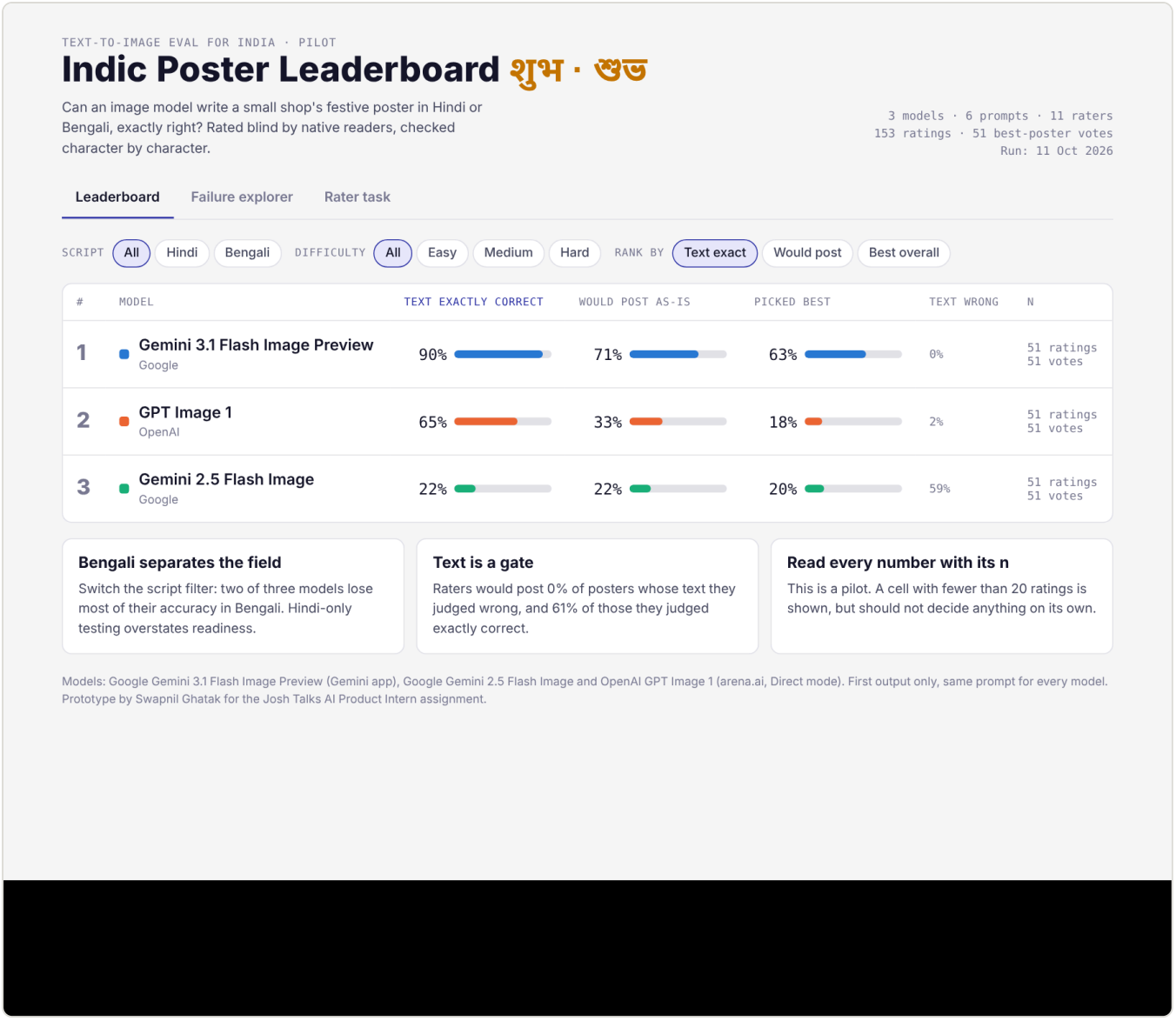


Figure 4. Leaderboard view, with filters and sample sizes.

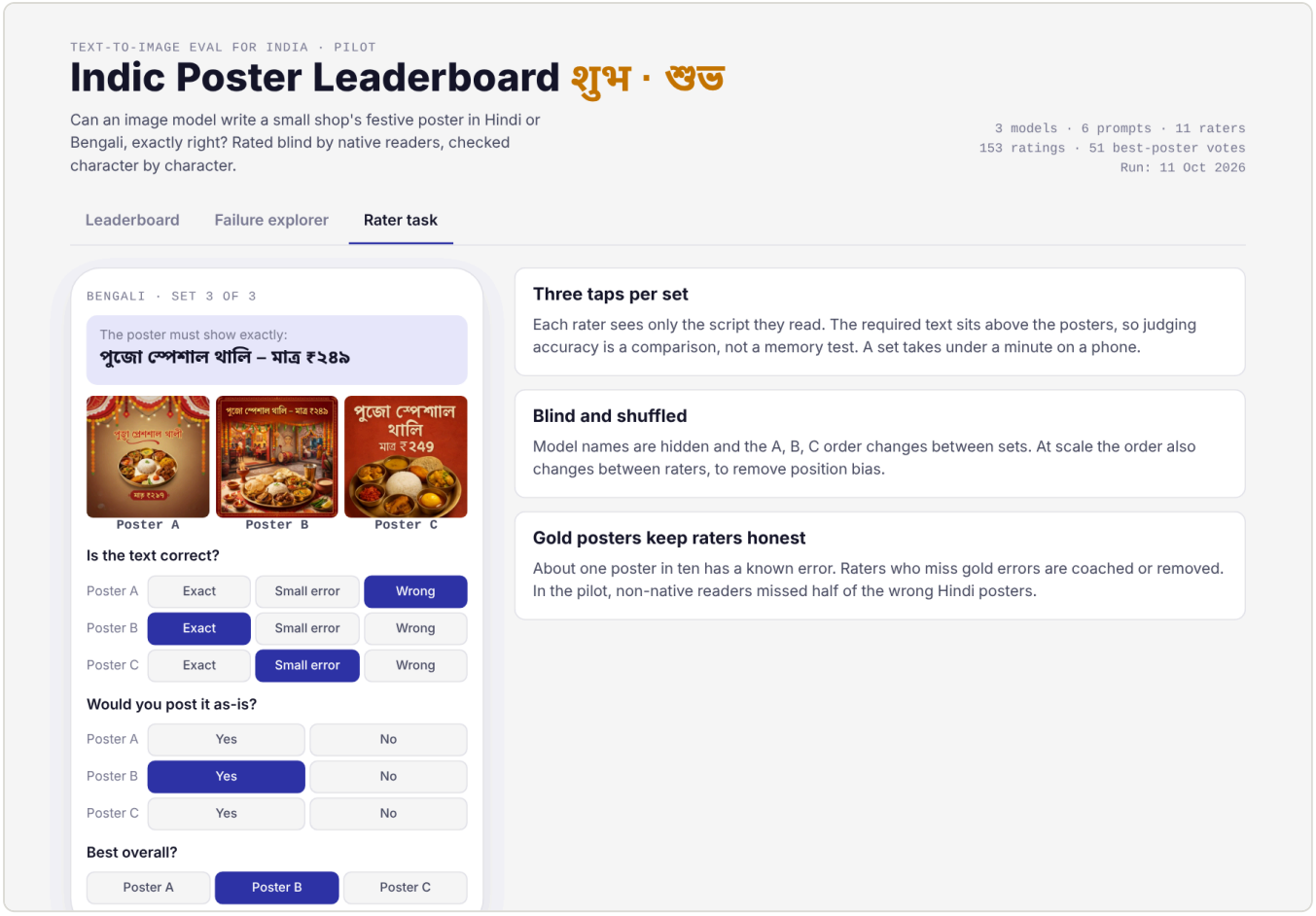


Figure 5. Rating screen: the required text above three unnamed posters.

6. Final reflection

Why did you choose this eval?
I care about how AI works for Indian languages, so I wanted an image test where language is the main thing being tested. Text inside images is a known weak spot for these models, and you can check it objectively, letter by letter, so the results are hard to argue with. I also read Hindi and Bengali, so I could check every poster myself and see how reliable the raters were.

Why is it useful for India?
There are crores of small businesses whose customers read regional scripts, and many of them advertise on WhatsApp and Instagram, especially around festivals. A tool that gets their poster right saves them paying a designer. One that misspells the greeting or changes the price costs them customers' trust and money.

Why would an AI lab building for India care?
Because testing only in Hindi hides the problem. Two of the three models lost most of their accuracy in Bengali. This test also shows which characters break (vowel signs, conjuncts, nukta, native digits), which tells a lab what data to add or what to fine-tune. And it connects accuracy to something real: would someone actually post this?

What did you learn from running the sample?
Correct text is the bare minimum. Nobody would post a poster with wrong text, but correct text alone didn't make people prefer a poster. I also learned that the raters matter as much as the models: people reading Hindi as a second script accepted about half of the wrong Hindi posters, and small mistakes got past almost everyone. On the practical side, my first form was too long. Cutting it to about 4 minutes, and showing each person only their own script, is what got me 11 responses in an afternoon.

What would you improve with more time?
More scripts (Tamil, Marathi, Odia, Gujarati and others) and more prompts at each difficulty level; several

images per prompt to see how consistent each model is; using the APIs so settings are fixed; shuffling A/B/C for each rater; hidden test posters to check raters; more real shop owners; an OCR first pass; and preference scores with confidence ranges on a live per-script leaderboard.

Appendix A. Participants

Everyone is 18 or older and ticked a required consent statement agreeing to their responses being used. To protect their privacy, names and emails are not published here; raters are shown by ID. My own test response is not included.

Rater	Rated	Runs a shop/business	Consent given
R01	Hindi	No	Yes
R02	Hindi, Bengali	No	Yes
R04	Hindi, Bengali	Yes	Yes
R05	Hindi, Bengali	No	Yes
R06	Bengali	No	Yes
R07	Hindi	No	Yes
R08	Hindi	Yes	Yes
R09	Hindi, Bengali	Yes	Yes
R10	Hindi, Bengali	No	Yes
R11	Hindi	No	Yes
R12	Hindi, Bengali	No	Yes

Appendix B. All 18 posters and my check

	Gemini 3.1 Flash Image Preview	GPT Image 1	Gemini 2.5 Flash Image
P01 Hindi Easy	 Exact: Correct	 Exact: Correct	 Wrong: दीपावली → दोपावली; adds unrequested text
P02 Hindi Medium	 Exact: Correct	 Exact: Correct	 Wrong: त्योहार → त्येहार, पेशकश → पेबस्शक
P03 Hindi Hard	 Exact: Correct, both lines	 Minor: सिर्फ़ written without nukta; odd line break	 Wrong: मोबाइल → मोबाल, शुरू → सुण, संपर्क → संपक
P05 Bengali Easy	 Exact: Correct	 Exact: Correct	 Wrong: শুভ → শুব; malformed glyphs

P06
Bengali
Medium



Exact: Correct



Minor: বিশেষ → বিশেব (ষ replaced by ব)



Wrong: পুজার শিশ্রস ছার্দ (unreadable)

P07
Bengali
Hard



Exact: Correct, incl. Bengali numerals



Minor: Price in Latin digits (249), not Bengali (২৪৯)



Wrong: থালি → খালী, price ₹২৪৯ → ₹২১৭

Appendix C. Exact prompts

ID	Prompt (same for all three models)
P01	Create an image: A square promotional poster for a traditional sweet shop in Varanasi for Diwali. The poster must display exactly this text in Hindi (Devanagari script), spelled correctly: "शुभ दीपावली". Warm festive style with diyas and sweets. No other text.
P02	Create an image: A square promotional poster for a saree shop in Lucknow for the festive season. The poster must display exactly this text in Hindi (Devanagari script), spelled correctly: "त्योहार की ख़ास पेशकश". Elegant style with colourful sarees. No other text.
P03	Create an image: A square promotional poster for a small mobile repair shop in Patna. The poster must display exactly this text in Hindi (Devanagari script), spelled correctly, as two lines: "मोबाइल रिपेयर – सिर्फ़ ₹199 से शुरू" and "संपर्क करें: 98765 43210". Bold, modern style. No other text.
P05	Create an image: A square promotional poster for a traditional sweet shop in Kolkata for Durga Puja. The poster must display exactly this text in Bengali script, spelled correctly: "শুভ শারদীয়া". Festive style with Durga Puja motifs and sweets. No other text.
P06	Create an image: A square promotional poster for a saree shop in Kolkata for Durga Puja. The poster must display exactly this text in Bengali script, spelled correctly: "পুজোর বিশেষ ছাড়". Elegant style with colourful sarees. No other text.
P07	Create an image: A square promotional poster for a small restaurant in Siliguri for Durga Puja. The poster must display exactly this text in Bengali script with Bengali numerals, spelled correctly: "পুজো স্পেশাল থালি – মাত্র ₹২৪৯". Appetising style showing a Bengali thali. No other text.

MSME figure: IBEF, "Over 7.83 crore enterprises registered on Udyam platforms", March 2026. Gemini app model: Google identifies Nano Banana 2 as Gemini 3.1 Flash Image.